Starting of Quick Sort

[Take first Element = pivot]

i:] 157, 110, 107, 122, 111, 149, 151, 141, 123, 112, 117, 133

Step:-1 pivot = 157

↳ All elements are less than 157,
↳ So after partiopation

110, 107, 122, 111, 149, 151, 141, 123, 112, 117, 133 / 157

Step:- 2 pivot = 110 (First element of left subarray)

107 / 110 / 122 111 149 151 141 123 112 117 113 / 157

Step:-3 Pivot = 122

partition the right subarray of 110

107 / 110 / 111 112 117 / 122 / 149 151 141 123 133 / 157

Step:-4 Pivot = 149

partition remaining right part

107 110 111 112 117 122 123 133 141 / 149 / 151 157

Final sorted array (after 4 steps)

107 110 111 112 117 122 123 133 141 149 151 157

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Take Last element = Pivot element

Step:-1

pivot = 133 [Last element]

partition:-

Left (<133): 110 107 122 111 123 112 117

Right (>133): 157, 149 151 141

110 107 122 111 123 112 117 | 133 | 157 149 151 141

Step:-2

pivot = 117 (Last of left subarray)

Partition left part:

110 107 111 112 | 117 | 122 123 | 133 | 157 149 151

141

Step:-3

pivot = 141 (Last of right subarray)

partition right part.

110 107 111 112 | 117 | 122 123 | 133 | 141 | 157

149 151

Step:-4

pivot = 151

partition remaining right part:

110 107 111 112 | 117 | 122 123 | 133 | 141 | 149

151 | 157

Final Sorted Array:-

107 110 111 112 117 122 123 133 141

149 151 157

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Take element = Random

I am taking the Random element = 149

Step:-1

pivot = 149

partition:

Left (<149): 110 107 122 111 141 123 112 117 133

Right (>149): 157 151

110 107 122 111 141 123 112 117 133 | 149 | 157 151

Step:-2

pivot = 141 (Middle of the left sub array)

110 107 122 111 123 112 117 133 | 141 | 149 | 157 151

Step:-3

pivot = 117 (Middle of the left subarray)

110 107 111 112 | 117 | 122 123 133 | 141 | 149 157 151

Step:-4

pivot = 151 (Middle of the right subarray)

110 107 | 111 112 | 117 | 122 123 133 | 141 | 149 | 151 157

Final Sorted array

107 110 111 112 117 122 123 133 141 149 151 157

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